Bridgewater Crossing (former Crain Brothers site) at the point of the Beaver and Ohio Rivers is being proposed. This regional attraction will be both a public park and a mixed use development.

Fallston is the home of the Beaver Valley Yacht Club on the Beaver River, which provides marina services for its members and limited restaurant services to the public.

Freedom Point Park has open space for picnics, and fishing access on the Ohio River.

Greene Township is proposing a rustic riverfront park on the Ohio River connecting into a 56 acre recreational park in the Township.

Monaca's riverfront along the Ohio River offers a public boat launch and river-viewing areas. The proposed rehabilitation of the Water Works public park site will provide playground equipment and proposed trail linkages to the community and to historic sites.

New Brighton Borough riverfront along the Beaver River offers marinas, a public boat launch, Big Rock Park, river access for fishing and viewing, and park benches. The New Brighton Water Works site is also proposed for a passive-use public fishing park.

Patterson Township has the potential to create a scenic overlook of the Beaver River Valley.

Rochester Borough riverfronts along the Beaver and Ohio Rivers contain a Riverfront Park, public playground, riverfront walkway, picnic shelters, boat launch and wharf, and the County's Flag Plaza. This regional attraction will be enhanced with proposed mixed use development. Recreational excursion vessels such as the Gateway Clipper and vessels housing cultural exhibits are accessed from the wharf.

Rochester Township riverfront along the Beaver River is proposing a riverwalk, and a (limited) river access area north of McKinley Run.

Site Analysis

Utilizing a detailed topographic and boundary survey, an analysis was conducted of the site for the proposed Beaver Borough Riverfront Park. This analysis, as it relates to the suitability for the park's intended recreational use, is summarized below.

Location and Size

The site is located in the southeastern corner of the town of Beaver, along the Ohio River just west of the juncture with the Beaver River. Approximately twelve acres in size, the pie shaped parcel is zoned P-1, Public, in the Zoning Ordinance of the Borough of Beaver. The purpose of this district is to preserve and protect land for open space and public recreation. Municipal parks, facilities, and services are permitted in this district.

Adjacent Land Uses

Riverview Park, also zoned P-1, borders the length of River Road above the site. Wayne Park is located opposite the site entrance at Beaver Street. Single-family residential (R-1) borders the remaining portion of River Road. The Ohio River borders the entire southern side, with the wastewater treatment plant to the east (across the Bridgewater-Beaver Borough boundary). Norfolk Southern Corporation's railroad tracks border the Northern side.

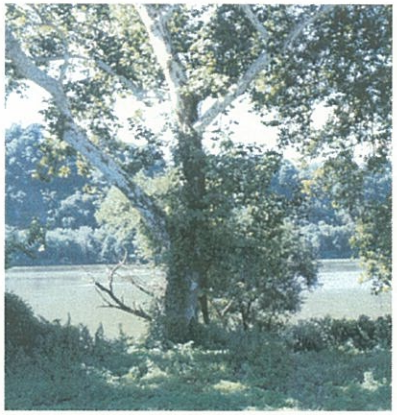
Adjacent to the wastewater treatment plant is a massive railroad bridge, which forms a physical and visual separation between the site and the parcel of land at the juncture of the Ohio and Beaver Rivers. This parcel was formerly home to the barge repair and dredging operation of Crain Brothers, Incorporated. In September of 1999, the Beaver County Corporation for Economic Development acquired the property. The parcel is now referred to as Bridgewater Crossing.

Site Ownership and Boundary

The site parcel is owned by the Beaver Borough Municipal Authority. The adjacent parcel in Bridgewater Borough, in which the wastewater treatment plant is located, is also owned by the Municipal Authority. The Municipal Authority, appointed by Beaver Borough Council, agreed to cooperate with the Borough toward the development of the site based upon the Master Plan.

A 33'-0 unopened lane, or paper street, bisects the site. This was vacated by the Borough at the turn of the century, and is now also owned by the Authority. Norfolk Southern Corporation owns the land occupied by its railroad tracks (approximately a one-hundred foot right-of-way), which forms the northern border.

The site is comprised of three parcels, two of which are adjoining, and the third of which is bisected by the 33'-0 paper street. Deed information for each of the three parcels, plus the paper street, was obtained from Deed Book Volume 1230, page 557, 10-9-1984.



The southern boundary at the Ohio River is the fluctuating normal (low) pool elevation at 682'-0. For all intents and purposes, this represents the waterline wherever it may be at any given moment. The area between the low and high water marks are subject to easement in favor of the public, except as required for transportation or channel improvement.

Topography and Vegetation

The majority of this riparian land is relatively flat (less than five percent slopes), with open, mowed field areas. A steep bank with grades greater than 25% separates the tracks from the town of Beaver, to the north, above. The water treatment plant is situated in fill areas with side slopes in excess of 25%. A sizeable depression occurs towards the eastern side of the site. This is where the majority of wildlife are present on the site including rabbit, pheasant, groundhog, deer, turkey, and numerous birds. Banks of varying grades, formed by wave action, line the river's edge.

Mature trees, representing species such as sycamore, box elder, willow, locust, and silver maple are thriving at the river's edge. A narrow wooded area, consisting mainly of box elder, separates the site's level portion from the railroad tracks. Brush vegetation of native and non-native field grasses and shrubs are understory to these vegetated areas.

Water Features

The Ohio River was formed by the confluence of the Allegheny and Monongahela rivers. Being over nine hundred miles long, it is a major tributary of the Mississippi River, and supplies more water than does the Missouri River. The Ohio River basin, in which the Riverfront park site is situated, covers over two hundred thousand square miles.

The site is located between river mile 26 and 27 along the Ohio River. The survey, benchmark from which topographic elevations are referenced, was obtained from Benchmark RM 2 on the Flood Insurance Rate Map.

The majority of the site is within the one hundred year flood plain, at or below elevation 703.00 feet. Normal (low) pool elevation is 682 feet, and defines the boundaries of land ownership for the parcel. High flood water elevation is 696.50 feet at 10 year, 701.50 feet at 50 year, and 708.00 feet at 500 year, with the floodway confined basically to the bank of the River. The sizeable depression on the site appears to drain freely after a storm event, resulting in little or no residual ponding of water.

Soils

Based on the Soil Survey for Beaver and Lawrence Counties, completed in 1977, the site soil is Philo Silt Loam. The survey also depicts a small water area in the northeast corner, which may have represented the frog pond fondly referred to by older residents. This water does not exist today. Although a wetland delineation was not performed under this study, a naturalist reviewed the area. Other than at the edge of the river, the site did not appear to contain jurisdictional wetlands.

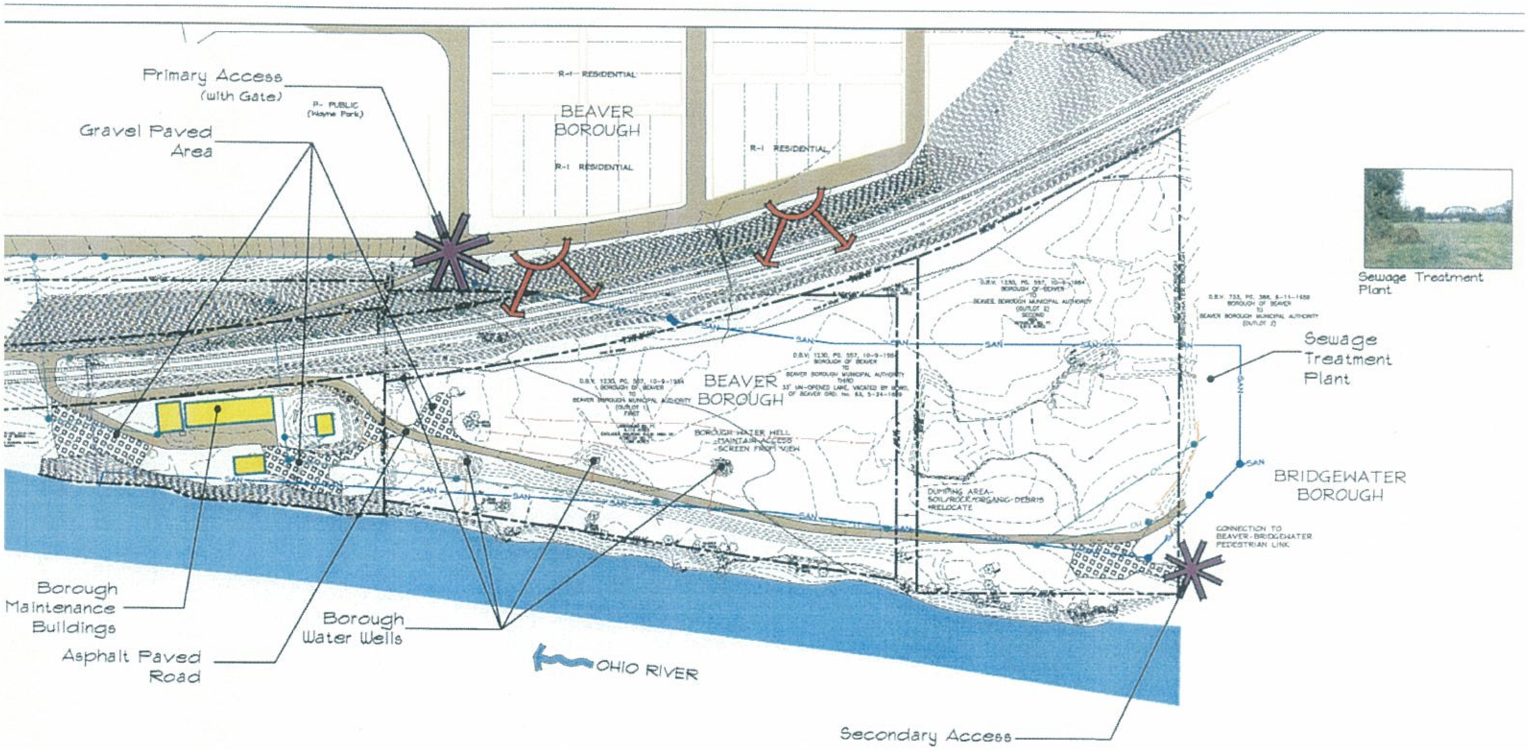
Deposits of soil and fills of varying sources have been placed on the site over the years. The water treatment plant appears to be situated on fill soils, above flood elevation.

Philo Silt Loam is characterized by deep, moderately drained silt loam soils with a moderate permeability (drainability of the soil), high available water capacity (water available to plants), and a seasonal high water table. These soils are generally found on bottomlands. Philo Silt Loam qualifies as prime farmland, making it suitable for trees and other plantings tolerant of wet conditions.

Philo soil is listed as a mapping unit with inclusions of hydric components (Atkins and Holly). Hydric soils are one of three factors of

Site Analysis of Man-made Features

*Refer to attached large scale drawing for additional information



Sewage Treatment Plant

Not to Scale


Water Treatment Plant


Maintenance Building


Gravel Storage


Railroad Crossing


Water Well

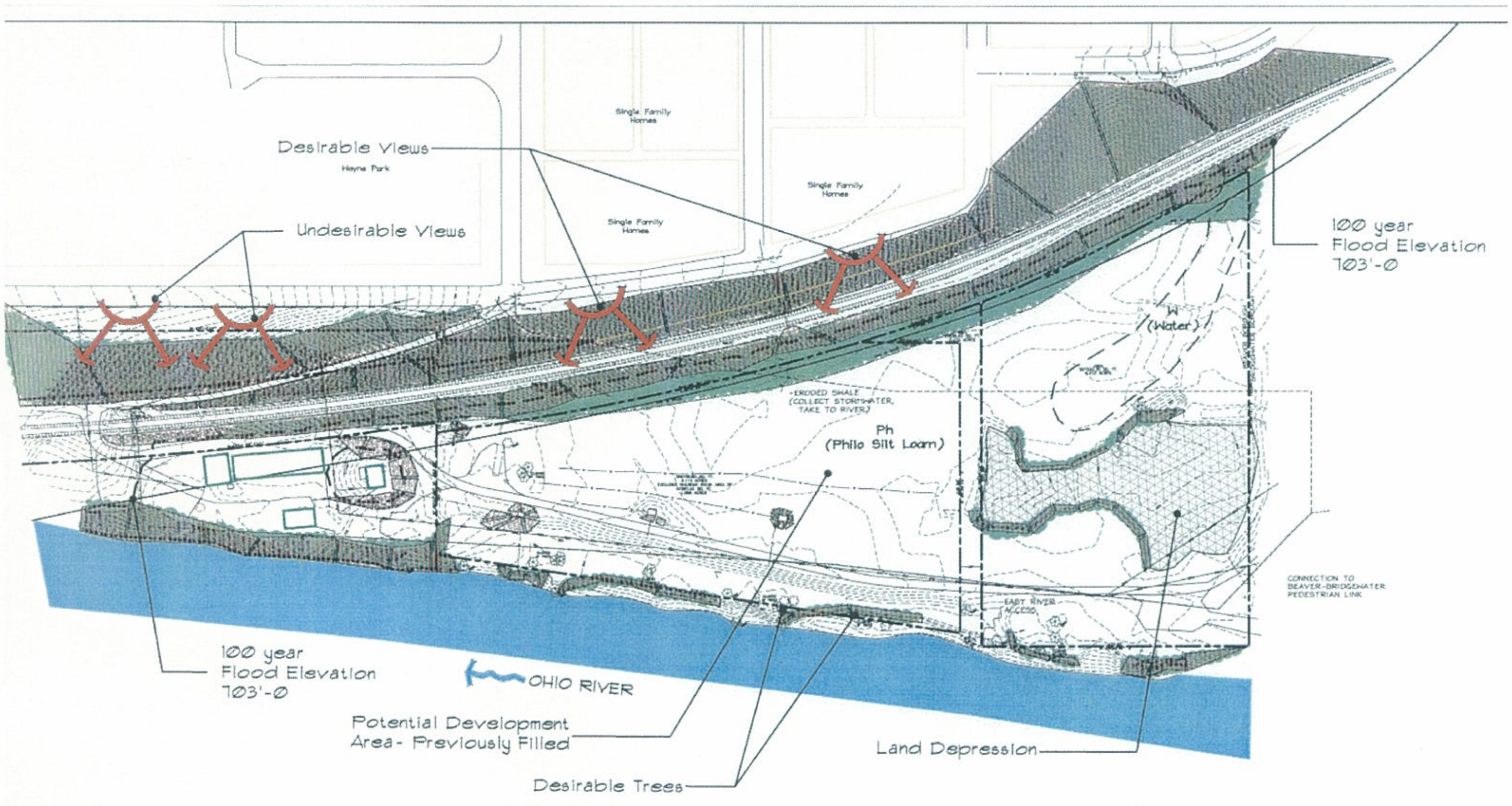

Water Supply Area


Vehicular Access


Maintenance Facilities

Site Analysis of Natural Features

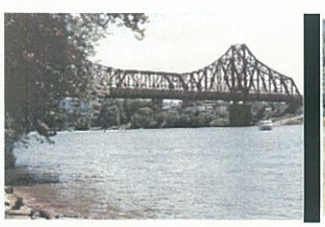
*Refer to attached large scale drawing for additional information




View of River


Distant Views

Not to Scale


Distant Views


Developable Area


Trees at River Edge


Water Supply Area


Grade Down To River

jurisdictional wetlands, with the other two being hydrophytic vegetation and hydrology.

Buildings

Borough maintenance facilities on the site include two storage sheds, one salt storage shed, and laydown space for equipment and gravel. A small water treatment plant, utilized primarily for water testing, is located near the storage sheds. A chain-link fence encloses a portion of these maintenance facilities.

Roads

Site access is accomplished via one-way, asphalt paved service roads from River Road and from adjacent properties in Bridgewater Borough. From Beaver Borough, the entering one-way road intersects River Road at Beaver Street, while the exiting one-way road intersects River Road at Market Street. Site access is restricted between the hours of 8:30 p.m. and 7:00 a.m, mainly to curb nighttime dumping.

At the downhill convergence of these one-way roads is a public road crossing of Norfolk Southern Corporation's railroad tracks. Current usage averages four to five trains per day which stop, on occasion, while cuing into Conway Yards. This results in the public crossing being blocked periodically.

An awkward left turn is required after crossing the tracks. The paved road then continues past the maintenance facilities, across the site, and accesses the wastewater treatment plant and the Borough's resident yard waste area. The road then passes under a railroad bridge, connecting the site to Bridgewater Borough.

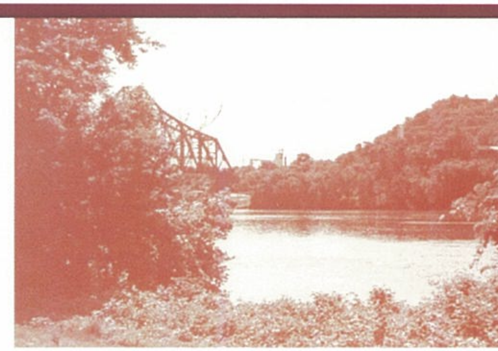
Utilities

Four active water wells exist in random locations on the site, with subsurface water lines of varying capacities. The well pumps access an Artesian water source well below the bed of the Ohio River. At each well site, soil is mounded up to the top of the well cover, allowing ease of service access. In addition, a water supply area is utilized for sale of water. Subsurface sanitary sewer lines occur, which connect the town's sanitary lines to the treatment plant and back to the outflow point on the riverbank.

Overhead electric lines follow the one-way service roads, crossing the tracks and connecting to the maintenance facilities. The overhead lines also traverse the site access road, serving the wastewater treatment plant. Portions of abandoned overhead lines remain along the railroad tracks. Utilities with lines at or serving the site include Dominion Peoples (gas); Columbia Gas of Pennsylvania, Incorporated (gas); Qwest Communications Corporation (communications); Verizon Pennsylvania Company (communications); and Beaver Borough Municipal Authority (water and sanitary sewer). Adelphia Cable and Duquesne Light Company have no utilities within the site. Addresses for the utility companies can be found in the Report Reference List.

Chapter 2

Public Participation



Chapter 2

Public Participation

This chapter describes the public participation process and summarizes results from each venue of participation. The input culminates in the identification of vision elements, leading to a "vision" for development of the Beaver Borough Riverfront Park.

The public input process included public meetings, working sessions with the project study committee, key person interviews, and meetings with student representatives of the Beaver Area School District.

To maximize attendance at public meetings, flyers were delivered to each address in Beaver and posted in prominent locations throughout the town. The meetings were also mentioned in the community newsletter and in the Beaver County Times. In addition, other agencies notified included the Beaver Area Chamber of Commerce, the Beaver Business District Authority, the Beaver County Riverfront Advisory Committee, the Beaver Initiative for Growth Recreation Committee, and the Beaver Rails to Trails Association.

Public Meetings

Three public meetings were held over the course of the study. The intent was to gather input and support, ensuring the recommended park improvements were consistent with the needs of the community. Minutes of each public meeting can be found in the Appendix.

Public Meeting #1

Sixty-two people attended the initial public meeting at the Beaver Area High School on November 8, 2000. A tailored input process was utilized to elicit ideas. Three questions were posed, and attendees were asked to write down their answers or comments:

1. What would you like us to know about the site?
2. What types of facilities or programs would you like to see incorporated in the design, serving what age groups?
3. What other issues, ideas or opportunities do you see for the park?

In consecutive order of seating, each attendee expressed only one idea at a time. This method assured all attendees had the opportunity to speak. After all ideas were recorded and posted, attendees prioritized five elements or ideas most important to them.

Slides of the site were shown at the first two public meetings. This reminded attendees of the character and the context of the site for which they were expressing ideas.

Key Issues Prioritized at Public Meeting #1

- Youth recreation area. Address skateboarding and rollerblading.
- Safe access. Consider pedestrian access over tracks and secondary access from Bridgewater.

The public input process was a key component in the development of recommendations for the park.

- Wildlife habitat. Respect bird nesting and migration and create wildflower areas.
- Trails. Provide walking and biking trails for exercise, with fitness stations.
- Simple design. Provide access to river, such as a boardwalk, fishing platform, sandy beach, boat access, and sitting areas.
- Yard waste area. Maintain resident yard waste area.
- Ice skating. Provide area for winter use.
- Picnic tables and shelters. Provide a place to picnic.
- Gardens. Plant ornamental flowers and provide areas for community gardens
- Cultural attraction. Consider a cultural areas and activities.
- Limited car access. Minimize impact of cars.
- Connectivity. Connect park trails into regional trail system.

Support facilities identified were drinking fountains, emergency phones, first aid and life preservers, trash receptacles, and a restroom.

Public Meeting #2

Seventy-one people attended the second public meeting held at the Beaver Area High School on April 18, 2001. Three schematic design alternatives were discussed and opportunities and constraints identified. Elements in the design were based on input received from the public input process performed to date. Main differences in the alternative schematics pertained to site access, as the status of the railroad crossing was unknown at the time they were developed.

Next, the preliminary design was discussed. The design incorporated positive elements from each of the three schematic designs. Main site access would remain over the existing track crossing, verified by Norfolk Southern Corporation as being a public crossing.

A lively discussion ensued. Comments were collected and discussed at the next study committee meeting.

Public Meeting #3

Thirty-seven people attended the third and final public meeting held at the Beaver Borough community room on October 30, 2001. A brief summary of teh mast planning process was reviewed. The recommended Design was explained in detail, including estimated costs, phasing, and funding opportunities.

The plan was well-received by the meeting attendees.

Student Meetings

An input meeting was held with members of the Beaver Area High School student council to gather ideas for the park. Note: the key issues summarized herein were not prioritized by the students.

Prior to the meeting, the recreation director for the school district conducted an independent recreation survey with school students to determine their needs and desires. The results of this survey can be found in the Appendix.

Key Elements Identified By Student Council

- Gathering spot. A place to meet with friends to talk and play music; an outdoor amphitheater (for bands to play and students to perform) or a big movie screen with the river as the backdrop; an indoor hang-out facility with recreation, refreshments, and climbing wall; a space for community gatherings; a bonfire area.
- Capture natural beauty. The former frog pond recreated.
- Recreation for all ages. Skateboard area; ice skating rink that can be converted for rollerblading; sand volleyball courts; river swimming (is water clean enough?); High school age challenge equipment or playground structure.
- Focal point. Monument or peaceful and serene water element.
- Access. Pedestrian and boater access. A spot for fishing.
- Trails. Walking, running and bicycle trails.
- Themed restaurant. Accessed from boats (maybe on a floating barge), or a snack bar in a lighthouse.
- Picnic areas.
- Flower gardens. A place to plant and enjoy the beauty.

Students expressed concern over the potential smell from the wastewater treatment plant. More importantly, they were adamant about limiting emphasis on environmental education. They felt there was too much "schooling" during the week.

Key Person Interviews

Key persons, identified by the study committee, were interviewed via phone to gather input for the project. The purpose of key person interviews was to give community leaders (beyond the elected officials) an opportunity to voice their concerns and be engaged in the planning process.

Key Issues Identified Through Key Person Interviews

1. What recreation opportunities would you like to see along the Beaver riverfront?

   - Park-like atmosphere
   - Multi-use space, with active and passive recreation
   - Walking access to Beaver Area Historical Museum

2. What key issues are facing the Borough in providing recreation opportunities for Beaver Borough residents?

   - Flooding
   - Access and parking
   - Railroad crossing safety
   - Hillside erosion
   - Funding
   - Vandalism
   - Sharing recreation resources (with neighboring communities)

3. What age groups appear to be the least / best served in Beaver Borough?

   - Best served: younger children (K-9); middle-aged (30-35); seniors
   - Least served: high school (no recreation opportunities unless school related); junior high (pre-teens to teens); middle aged; young adult (early 20's to 30's).

4. What kind of special events / programs would you like to see occur at the riverfront?

   - River regattas, water sports, fishing derbies, and events (jet skis, crew racing)
   - Outdoor music events and amphitheater
     - Community and civic events
     - Ecological activities
     - Rock climbing programs
     - Community gardens
     - Historical programs (related to Fort McIntosh)
     - Nature trails and programs

Study Committee Working Sessions

A ten-member study committee was instrumental in developing the Beaver Borough Riverfront Park Master Site Development Plan. The committee included representatives from Beaver Borough Council, public works, planning and zoning, public safety, and the Borough manager, as well as the School District recreation director, and Beaver residents. Minutes from the three meetings held with the study committee can be found in the Appendix.

The role of the study committee was to:

a. Act as a sounding board for ideas generated through the public input process.
b. Provide feedback on design alternatives.
c. Comment on recommendations.

Key Elements Identified by the Study Committee

- Sports fields
- Boardwalk
- Amphitheater
- Unique feature or attraction (lighthouse)
- Bicycle and walking trails (connections to Bridgewater)
- Ecological development, with natural theme
- Environmental education (with School District)
- Beach
- Water feature
- Boat access
- Playground for all ages
- Ice rink
- Family picnic areas
- Fitness trail with stations
- Fishing

Other issues identified as important:

- Emphasize year-round use, acknowledging 100 year flood elevations.
- Provide boater access, with tie-up points (rather than boat docking).
- Lower top elevations of

existing sanitary sewer manholes to accommodate improvements. Current top elevations are not above 100 year flood elevation.
- Gate park access points to curb nighttime dumping.
- Address, on a County-wide basis, active recreational uses requiring parking and lighting, such as ballfields, and skate parks.
- Address concern over potential vandalism given the site's location. Frequency of visitors on a random basis will help curb vandalism.
- Determine cone of depression around each well source, per the Source Water Protection Program (as recommended by PADEP).
- Maintain current location of Borough maintenance facilities. Relocation of these facilities is beyond the scope of this study, however, the salt storage shed can be moved.
- Maintain access to Borough wells (for large crane truck), water treatment plant (plan for future expansion), and maintenance facilities.
- Maintain access to Borough dumping facility and resident yard waste area (consider off-site location).
- Include area for future expansion or modification of water treatment plant per anticipated future requirements of the Pennsylvania Department of Environmental Protection (PADEP).

Creating a Vision for the Future

How do residents of Beaver see their park years from now? To arrive at a "vision narrative" for the park, the consultant summarized the common ideas elicited from the public participation process.

Ideas, or "visions", consistently mentioned were grouped into vision elements and utilized to form the vision statement. The vision fulfills the desires of the community and illustrates the essence of how their park should appear sometime in the future. It provides a mental image which generates support for the project, and forms a basis for all future park development.

More active recreational facilities, such as ballfields and a skateboard park, were not included in the design for this site. After review by the study committee, with input from the consultant and the CED, these facilities were not considered compatible with a riparian environment. Ballfields would require more frequent fertilization. A skateboard park would require more extensive paved areas, and greater surveillance.

Concurrently, a feasibility study is being conducted to determine the support and need for a recreation center serving Beaver and its surrounding municipalities (Bridgewater, Vanport, and Brighton Township). In addition, a comprehensive park, recreation, and open space plan is being undertaken by Beaver County. Active recreation facilities will be considered with these studies to meet the needs of area residents on a regional basis.

VISION STATEMENT

- *Design a simple park with flexible passive recreation space offering amenities for all ages, while respecting the natural environment and maintaining access to existing Borough facilities.*

- *Create a park which serves as a natural destination for wildlife and human life communities, providing a haven where birds and animals are welcomed and observed as they establish or simply pass through.*

- *Provide a gathering place for conversation and a place people can go to be revitalized by the calming river environment.*

- *Establish a park which will serve as a restorative place for relaxing and reconnecting with the natural world. Encourage an educational experience gained through personal exploration.*

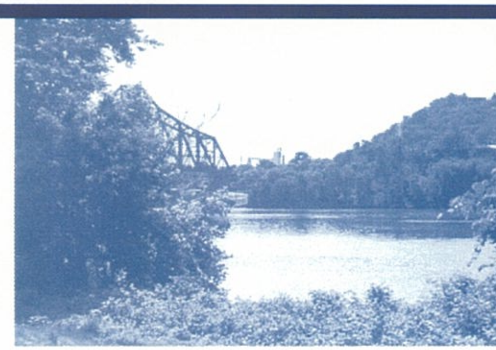
- *Create this park using natural materials and a simple flood resistant, and easy to maintain design.*

Chapter 3

Schematic Design Alternatives

Chapter 3

Schematic Design Alternatives



Schematic design alternatives were prepared for presentation at the second public input meeting. The design elements shown reflected thoughts and desires elicited from the first public meeting, with professional input from the consultant. They were prepared to express ideas, not a final design.

The Schematic Design Alternatives are briefly outlined below. Their opportunities are described by the elements included in each design. Constraints are listed at the end of each description. The concept for each was similar:

To create varying experiences for park visitors, a balance of open lawn area to accommodate pick-up games and community events, along with the extension of a natural habitat area, were designed. The features of the park will center around these two distinct areas.

Opportunities from each Schematic Design Alternative were incorporated in the Preliminary Design of the Beaver Borough Riverfront Master Site Development Plan.

Design Alternative #1
(access from River Road)

A large open lawn area was defined to accommodate pick-up games and community events. An irregular-shaped pond for ice skating was shown. Along side this open space, a continuous stone sitting wall with an adjacent path (rather than individual benches along a trail), would provide seating and a barrier from the railroad tracks. A wall would create a simple line, better withstand flooding, and require minimal maintenance.

An expansion of wildlife habitat was included to support a broader range of flora and fauna. Channeling of stormwater runoff through the habitat and providing nesting boxes would provide further support. Incorporating boulders and fence posts would facilitate wildlife viewing, and log benches would provide seating. A natural, curving wall made of local stone, apparently deposited during the Ice Age, would separate the natural habitat area. Educational signs would be inscribed in the wall.

An expansion of wildlife habitat was included to support a broader range of flora and fauna.

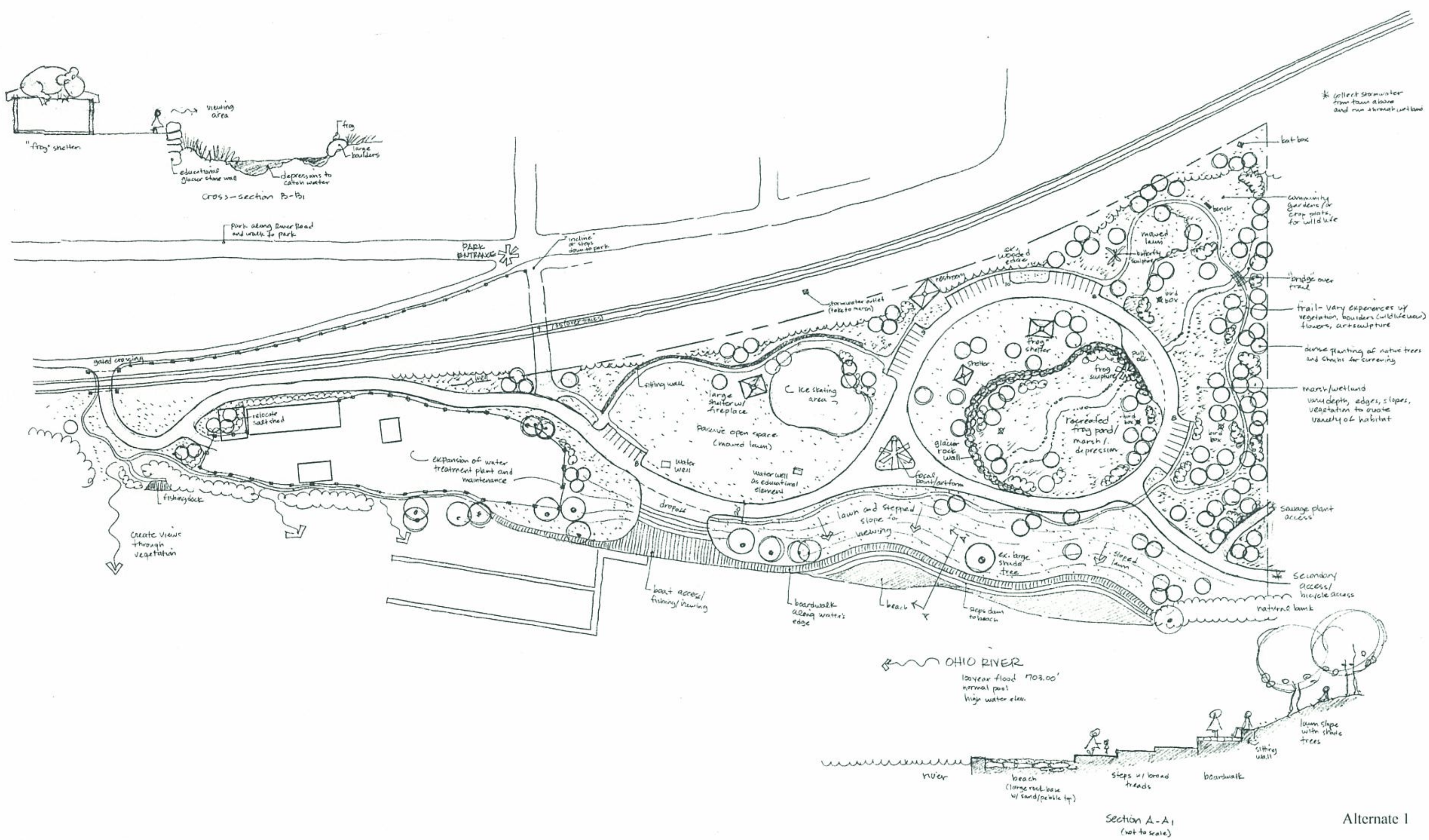
Picnic shelters were placed in lawn areas along the natural habitat area. They would be themed to acknowledge the site's former frog pond.

Multi-use trails were shown for walking and biking, with adjacent fitness stations. Nature trails located at the edges of the habitat area would reduce impact on wildlife.

Art sculptures, appropriate to the natural setting, were located within park. The community would be instrumental in their design and construction.

Community garden plots for vegetables and flowers were located within the natural habitat, to blend should their appearance become unkempt.

Design Alternative #1



Alternate 1

Riverfront viewing and sitting areas were created by terraced stone or concrete steps, along with extensive sloped mowed lawn areas. A boardwalk or river walk, located above the high water level to curb flood damage, would connect to Bridgewater Crossing. A concrete or stone edge along the river would be constructed for reduced susceptibility to water erosion. A sandy beach using a base of larger rock choked with a finer sand or pebbles would be constructed, also for reduced maintenance. Courtesy docking along the river's edge was shown to provide boater access

Storm water management from proposed improvements, and from the town of Beaver above, would be handled by on-site swales, allowing further filtering before entering the river.

Park signs of natural materials and colors would be limited in number, and include park entrance sign(s), accessible parking, park rules, and educational signage. One-way roads would be avoided to reduce sign clutter.

Pending a verification of a public track crossing, the existing park entrance, exit, and one-lane access roads would be enhanced with stone walls, entrance signs and gates, and attractive plantings. Safety would be improved through the installation of wooden guide rails.

The salt storage shed would be relocated to allow a safer road alignment, and a more aesthetic entrance. The site access road would loop, connecting to Bridgewater and maintaining access to the Borough's yard waste area. A focal point, such as a lighthouse, would announce the park to both land and water users. Reduced pavement widths would be utilized to minimize visual impact and reduce stormwater runoff. Forty-five parking spaces would serve picnic shelters, ice skating pond, and general park use.

Access to Borough water wells would be maintained, and consideration given to acknowledging the wells as public education tool describing the origination of resident's water. To reduce visual clutter, the overhead electric lines would be relocated to along the railraod tracks, or buried.

Native plantings (avoiding invasive species) for shade, and wildlife food and shelter, would be planted in random fashion, and clustered more closely along the edge to effectively screen the sewage treatment plant.

Alternate pedestrian access (steps or incline) was considered to avoid an at-grade crossing of the railroad tracks. It would not be feasible to meet ADA guidelines given the significant change in grade.

CONSTRAINTS

- Park road interrupts natural habitat area.
- More hard edges at riverbank.
- Multiple shelters shown but not desired.
- Boat docking facilities shown but not desired.

Design Alternative #2
(access from River Road)

A central plaza forms the main core of this alternative. Concentric "rings" emanating from this arrival space would take the form of sitting walls near the center, and become stone bands further out. These flush stone bands would cross lawn areas, trails, and roads. In the natural habitat, they would become rows of trees and remnants of fence posts.

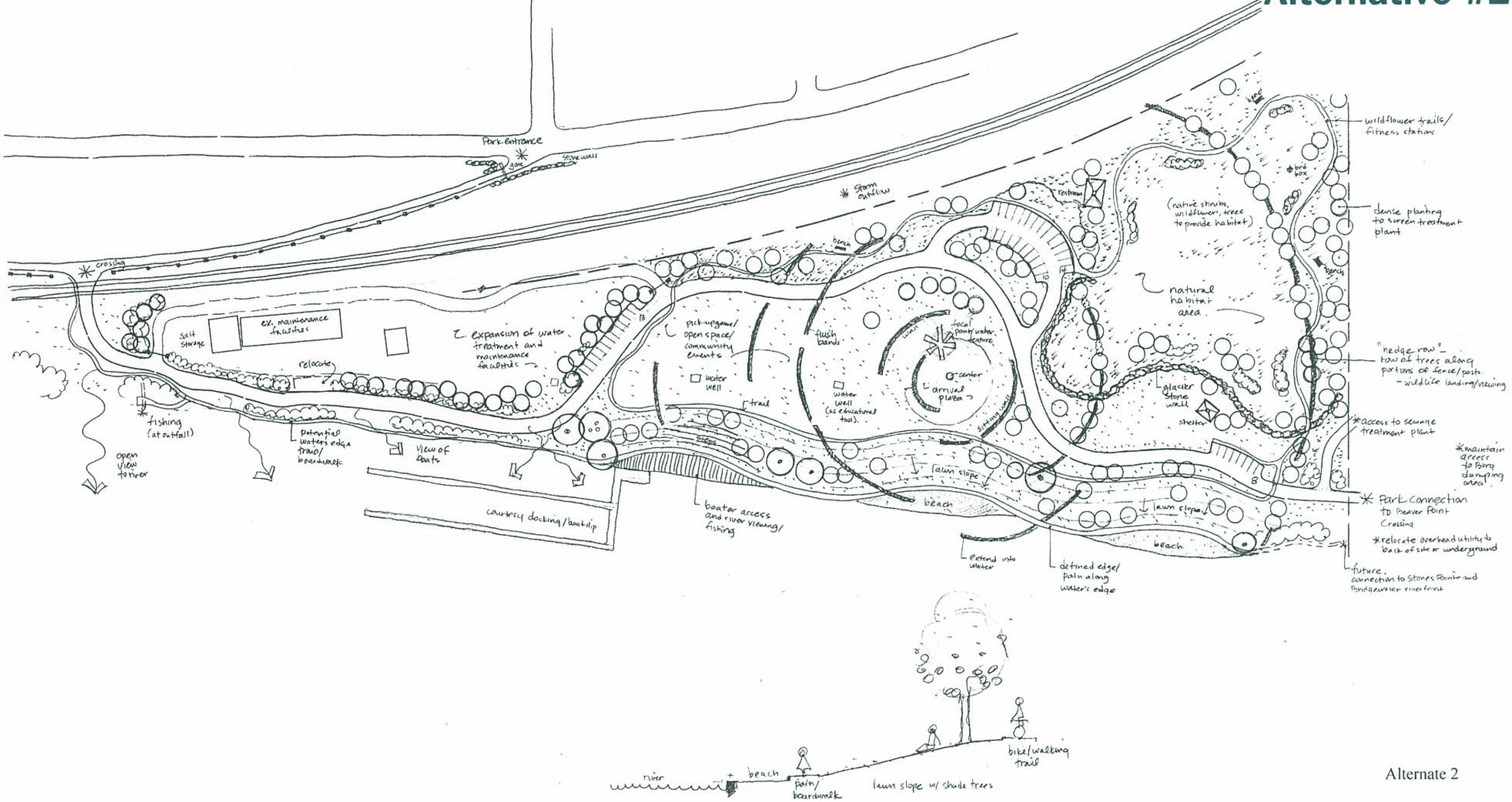
The main parking area was also curved to acknowledge the concentric theme. A picnic shelter would overlook the habitat, adjacent to a curving river stone wall.

Main park access would be from River Road, pending an approved track crossing from Norfolk Southern Corporation. The park road would then traverse along the riverbank, between the maintenance facilities and the top of the riverbank, requiring relocation of a maintenance shed. Forty-five parking spaces would serve picnic shelters, ice skating pond, and general park use.

CONSTRAINTS

- More hard edges at riverbank.
- Multiple shelters not desired.
- No looped road or turnaround provided in park, other than through parking area.
- Relocation of maintenance shed required to accommodate park road along river.
- Boat docking facilities shown but not desired.

Design Alternative #2



Alternate 2

Design Alternative #3
(access from Bridgewater Borough)

The third alternative was developed with the main park access from Bridgewater, assuming a public track crossing would not be granted from the railroad.

The natural habitat area would be located near the maintenance facilities, rather than adjacent to the wastewater treatment plant. A small water feature would be located. A more natural treatment along the riverbank was designed, and would consist of walking trails and native vegetation to further support wildlife.

The park access road would form a loop, with a central arrival plaza and a focal point. Picnic shelters would be located in an open lawn area, with lawn access to the river's edge. Thirty parking spaces would be constructed to serve picnic shelters and general park use.

CONSTRAINTS

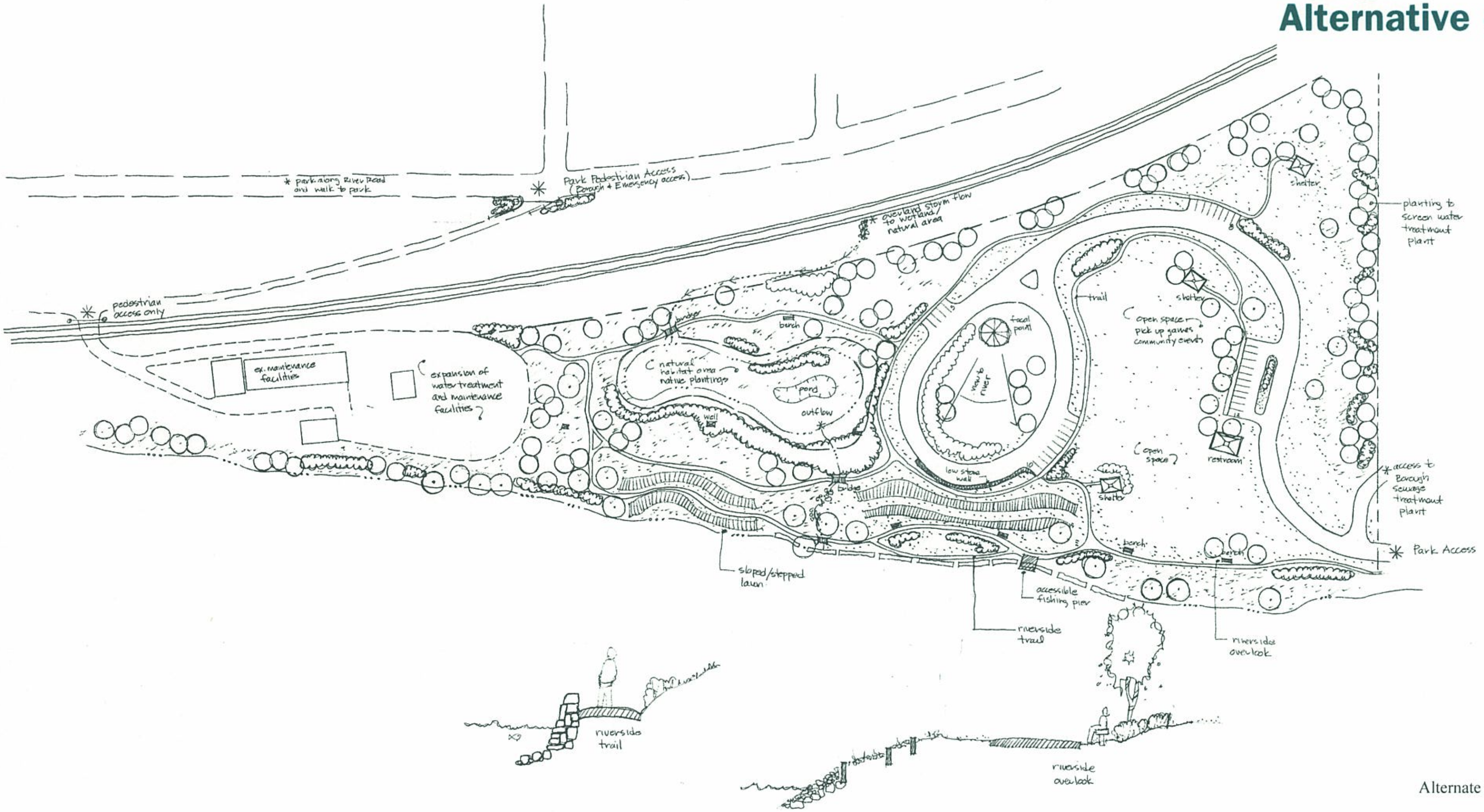
- Main park access from Bridgewater, not Beaver.
- Restroom not central to main activity areas.
- Multiple shelters shown but not desired.
- Natural habitat not contiguous with undeveloped area behind wastewater treatment plant.
- Less accessibility to riverbank for human visitors.
- No vehicular connection from park access road to Borough maintenance facilities.

Preliminary Design

The study group considered the schematic design alternatives, and discussed which elements should be included in the design of the Preliminary Master Site Development Plan for the Beaver Borough Riverfront Park. The group agreed that Design Alternative #1, with modifications, will best meet the community's needs. The following modifications were proposed:

- Maintain vehicular access to site via River Road (pending railroad company determination of public track crossing). Improve safety of track crossing by re-paving surface, installing cross-arms and signage, and other as coordinated with Norfolk Southern Corporation.
- Include an ice skating pond as a year-round, shallow pond, with a concrete base and possibly a fountain for circulation. Consider placing water lilies for summer interest, since cleaning and draining of the pond will be necessary prior to freezing for ice skating.
- Maintain evening gated access to site to curb illegal dumping. Borough may consider alternate yard waste removal for residents, rather than at the riverfront site.
- Create one large pavilion near the pond, with restrooms, fireplace, drinking fountain, and other amenities. Costs can be put into one, rather than several structures, allowing better construction materials (stone base and columns, with wood detailing and shingled roof overhead (above flood elevation to minimize damage). Locate shelter closer to river for greater use flexibility.
- Provide access to the shoreline for boaters, rather than courtesy docks.
- Designate a sizeable natural habitat area to allow diverse range of habitat, with natural stormwater drainage flowing through.

Design Alternative #3



Alternate 3